UCR University College Roosevelt

Fall 2026 cohort

# Astronomy at University of Groningen

Starting interests: **Space exploration and astronomy**

**Official programme ↗**

## Astronomy and astrophysics

- Introduction Astronomy · 5 EC
- Physics of Stars · 5 EC
- Physics of Galaxies · 5 EC
- Radiative Processes in Astrophysics · 5 EC
- Advanced Electrodynamics · 5 EC
- Astrophysical Hydrodynamics · 5 EC
- Interstellar Medium · 5 EC

## Physics and mathematics

- Mechanics and Relativity · 10 EC
- Electricity and Magnetism · 10 EC
- Thermal Physics · 10 EC
- Quantum Physics 1 · 5 EC
- Quantum Physics 2 · 5 EC
- Waves and Optics · 5 EC
- Calculus 1 (for Physics) · 5 EC
- Calculus 2 (for Physics) · 5 EC
- Linear Algebra (for Physics) · 5 EC
- Mathematical Physics · 5 EC
- Complex Analysis · 5 EC

## Computing, data and observation

- Introduction to Programming and Computational Methods · 5 EC
- Numerical Methods · 5 EC
- Statistics for Astronomy · 5 EC
- Observational Astronomy · 5 EC

## Exploration, instruments and responsible technology

- Minor: Instrumentation & Informatics · 30 EC

## Earth, planets and life

- Introduction to Planetary Science · 5 EC

## Research and academic development

- Physics Lab: Skills · 5 EC
- Astronomy Bachelor Research Project · 15 EC

UCR does not currently offer astronomy or physics courses. The UCR programmes shown here therefore build around the closest feasible combinations of mathematics, computing, observation, Earth systems, life science and technology.

Selected minor route: Instrumentation & Informatics (30 EC).

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# Computing, Data and Earth Observation

Starting interests: **Space exploration and astronomy**

**UCR courses ↗**

| Physics and mathematics |
|---|
| Advanced Calculus · 7.5 EC |
| Linear Algebra · 7.5 EC |
| Mathematical Methods · 7.5 EC |
| Logic & Discrete Mathematics · 7.5 EC |
| Linear Systems · 7.5 EC |
| **Computing, data and observation** |
| Introduction to Programming · 7.5 EC |
| Computational & Numerical Methods · 7.5 EC |
| Probability & Statistics · 7.5 EC |
| Earth Observation · 7.5 EC |
| Introduction to Data Science · 7.5 EC |
| Machine Learning · 7.5 EC |
| Image Processing & Computer Vision · 7.5 EC |
| Neural Networks & Deep Learning · 7.5 EC |
| Optimization Methods in AI · 7.5 EC |
| Reinforcement Learning · 7.5 EC |
| Algorithms & Data Structures · 7.5 EC |
| Networks & Operating Systems · 7.5 EC |
| Database Management Systems · 7.5 EC |
| Earth System Science Tools · 7.5 EC |
| **Exploration, instruments and responsible technology** |
| Robotics · 7.5 EC |
| Consumer Product Design · 7.5 EC |
| **Earth, planets and life** |
| Sustainability Essentials I · 7.5 EC |
| Environmental Biogeochemistry · 7.5 EC |
| **Research and academic development** |
| Personal & Professional Development · 7.5 EC |

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UCR does not offer astronomy or physics courses; this programme builds the closest feasible depth through mathematics, computing, data and observation.

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# Earth Observation, Robotics and Sustainable Technology

Starting interests: **Space exploration and astronomy**

**UCR courses ↗**

## Physics and mathematics

Linear Algebra · 7.5 EC

Mathematical Methods · 7.5 EC

Logic & Discrete Mathematics · 7.5 EC

## Computing, data and observation

Introduction to Programming · 7.5 EC

Computational & Numerical Methods · 7.5 EC

Probability & Statistics · 7.5 EC

Earth Observation · 7.5 EC

Introduction to Data Science · 7.5 EC

Machine Learning · 7.5 EC

Image Processing & Computer Vision · 7.5 EC

Optimization Methods in AI · 7.5 EC

Earth System Science Tools · 7.5 EC

## Exploration, instruments and responsible technology

Robotics · 7.5 EC

Consumer Product Design · 7.5 EC

Renewable System Design · 7.5 EC

AI Ethics · 7.5 EC

## Earth, planets and life

Sustainability Essentials I · 7.5 EC

Sustainability Essentials II · 7.5 EC

Dynamic Earth · 7.5 EC

Environmental Biogeochemistry · 7.5 EC

Earth's Resources · 7.5 EC

Climate Change · 7.5 EC

## Research and academic development

Research in a Sustainable Delta · 7.5 EC

Personal & Professional Development · 7.5 EC

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UCR does not offer astronomy or physics courses; this programme combines quantitative tools with Earth science and exploration technology.

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# Earth Systems, Technology and Life Sciences

Starting interests: **Space exploration and astronomy**

**UCR courses ↗**

## Physics and mathematics

Mathematical Methods · 7.5 EC

## Computing, data and observation

Introduction to Programming · 7.5 EC
Earth Observation · 7.5 EC
Introduction to Data Science · 7.5 EC
Machine Learning · 7.5 EC
Image Processing & Computer Vision · 7.5 EC
Neural Networks & Deep Learning · 7.5 EC
Earth System Science Tools · 7.5 EC

## Exploration, instruments and responsible technology

Robotics · 7.5 EC
Consumer Product Design · 7.5 EC
Renewable System Design · 7.5 EC
AI Ethics · 7.5 EC

## Earth, planets and life

Sustainability Essentials I · 7.5 EC
Sustainability Essentials II · 7.5 EC
Dynamic Earth · 7.5 EC
Environmental Biogeochemistry · 7.5 EC
Earth's Resources · 7.5 EC
Climate Change · 7.5 EC
Introduction to Biodiversity · 7.5 EC
The Science of Evolution · 7.5 EC
Foundation of Life, Health & Biomedicine · 7.5 EC
Biochemistry · 7.5 EC

## Research and academic development

Research in a Sustainable Delta · 7.5 EC
Personal & Professional Development · 7.5 EC

UCR does not currently offer astronomy or physics courses. The UCR programmes shown here therefore build around the closest feasible combinations of mathematics, computing, observation, Earth systems, life science and technology.

UCR does not offer astronomy or physics courses; this programme broadens into Earth systems, life science, data and exploration technology.